Dr.-Ing. Aurore Fass

Tenured Researcher at Inria

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Research Overview

My research work revolves around designing practical approaches to protect the security and privacy of Web users. I build systems to proactively detect malicious JavaScript code and suspicious browser extensions. I analyze data to understand how people spend time on the Web, and I want to use the resulting perspective to prioritize defense strategies.

Scientific Career

2025– **Tenured Researcher**, *Inria Centre at Université Côte d'Azur (Sophia Antipolis)*, France

2023–2025 **Tenure-Track Faculty (W2)**, *CISPA Helmholtz Center for Information Security*, Germany

2021–2023 **Visiting Assistant Professor**, *Stanford University*, U.S
- Host: Zakir Durumeric

2021 **Postdoctoral Researcher**, *CISPA Helmholtz Center for Information Security*, Germany

2017–2021 **Ph.D. Student**, *Saarland University & CISPA Helmholtz Center for Information Security*, Germany
- Ph.D. thesis: *Studying JavaScript Security Through Static Analysis*
- Advisors: Michael Backes and Ben Stock

Education

2014–2017 **Grande École** (similar to a Master Degree), *TELECOM Nancy*, France, **valedictorian**
Major: Telecommunication, Network, and Security
- Master thesis: German Federal Office for Information Security (BSI), Germany
  Automated clustering of JS samples for the detection of malware contained in obfuscated code
- Industrial project: French Ministry of Defense, France
  Implemented an Xposed module to monitor Android devices; group of 4 persons (6 months)
- Internship: Fraunhofer IOSB, Germany
  Implemented a passive asset detection system (8 weeks)

2012–2014 **Preparation for the highly competitive nationwide entrance examination to the French Grandes Écoles**, France
Major: Mathematics, Physics, and Computer Science

2012 **High school graduation**, France, graduated with distinction ("mention très bien"), European section
Major: Mathematics, Physics & Chemistry, Biology, and German

Awards and Honors

2025 **Distinguished Reviewer Award**, *ACSAC*

2025 **Top Reviewer Award**, *ACM CCS*

2025 **Distinguished Reviewer Award**, *USENIX Security*

2024 **Distinguished Reviewer Award**, *ACSAC*

2024 **Noteworthy Reviewer Recognition**, *EuroS&P*

2023 **Top Reviewer Award**, *ACSAC*

2023 **Top Reviewer Award**, *ACM CCS*

2022 **Best Reviewer Award**, *ACM CCS*

2022 **PC Member Honorable Mention**, *TheWebConf*

2021 **Inspiring Career Recognition**, *1 of 3 invited alumni (out of 2,300 alumni) for the 30th anniversary of the French Grande École TELECOM Nancy*, Remote

2019–2022 **Program of Excellence**, *Saarland University*, Germany

2017 **Valedictorian**, *French Grande École TELECOM Nancy*, France

2016 **Best Student Recognition Event**, *IBM*, UK

Funding and Research Grants

Since joining Inria in December 2025, I have acquired the following research grants:

2026–2030 **ANR JCJC (EXPELLIARMUS), Principal Investigator**

Competitive young researchers grant, funded by the French National Research Agency (ANR).

This project supports my research on disarming dangerous browser extensions to protect the security and privacy of Web users.

Duration: 42 months.

Total funding: **322,454€**.

2026–2029 **IdEx Attractivity Package**, **Principal Investigator**

Funding awarded to newly recruited researchers through a project-based selection process, within the Initiative of Excellence (IdEx) of Université Côte d'Azur.

This project supports my research on detecting dangerous browser extensions.

Duration: 3 years.

Total funding: **50,000€**.

Prior to joining Inria, I held a position with substantial institutional research funding, and therefore did not need to apply for external research grants. The grants listed above were obtained shortly after joining Inria, demonstrating my ability to successfully compete for external funding.

Publications

* Ben Rosenzweig, Valentino Dalla Valle, Giovanni Apruzzese, and **Aurore Fass**. It's not Easy: Applying Supervised Machine Learning to Detect Malicious Extensions in the Chrome Web Store. In *ACM Transactions on the Web (TWEB)*, 2026. Code repository: `https://github.com/its-not-easy/tweb25`.

Abdullah Hassan Chaudhry, Valentino Dalla Valle, and **Aurore Fass**. Two Heads are Better Than One: Analysing Browser Extensions Across Stores. In *MADWeb (workshop co-located with NDSS*, 2026. Code repository: `https://github.com/abdchau/two-heads-are-better`.

Pouneh Nikkhah Bahrami, **Aurore Fass**, and Zubair Shafiq. CookieGuard: Characterizing and Isolating the First-Party Cookie Jar. In *ACM Internet Measurement Conference (IMC)*, 2025. Code repository: `https://github.com/pooneh-nb/cookieGuard`.

Kimberly Ruth, Veronica Rivera, Gautam Akiwate, **Aurore Fass**, Kurt Thomas Patrick Gage Kelley, and Zakir Durumeric. 'Perfect is the Enemy of Good': The CISO's Role in Enterprise Security as a Business Enabler. In *ACM CHI*, 2025.

Saskia Laura Schröer, Giovanni Apruzzese, Soheil Human, Pavel Laskov, Hyrum S. Anderson, Edward W.N. Bernroider, **Aurore Fass**, Ben Nassi, Vera Rimmer, Fabio Roli, Samer Salam, Ashley Shen, Ali Sunyaev, Isabel Wagner, Gang Wang, and Tim Wadhwa-Brown. SoK: On the Offensive Potential of AI. In *IEEE Secure and Trustworthy Machine Learning Conference (SaTML)*, 2025.
Acceptance rate: 29.4% (53 / 180 full research papers).

Dominic Troppmann, **Aurore Fass**, and Cristian-Alexandru Staicu. Typed and Confused: Studying the Unexpected Dangers of Gradual Typing. In *IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2024. Code repository: `https://zenodo.org/records/13760256`.
Acceptance rate: 26% (155 / 587 full research papers).

Giovanni Apruzzese, **Aurore Fass**, and Fabio Pierazzi. When Adversarial Perturbations meet Concept Drift: an Exploratory Analysis on ML-NIDS. In *ACM AISec (CCS Workshop on Artificial Intelligence and Security)*, 2024. Code repository: `https://github.com/hihey54/aisec24`.
Acceptance rate: 25% (18 / 72 full research papers).

Shubham Agarwal, **Aurore Fass**, and Ben Stock. Peeking through the window: Fingerprinting Browser Extensions through Page-Visible Execution Traces and Interactions. In *ACM CCS*, 2024. Code repository: `https://github.com/raider-ext/raider`.
Acceptance rate: 18% (129 / 710 full research papers, Cycle A).

* Sheryl Hsu, Manda Tran, and **Aurore Fass**. What is in the Chrome Web Store? Investigating Security-Noteworthy Browser Extensions. In *ACM AsiaCCS*, 2024.
Media coverage: `https://aurore54f.github.io/papers/hsu2024cws.media`.
Acceptance rate: 22% (65 / 301 full research papers).

Liz Izhikevich, Manda Tran, Michalis Kallitsis, **Aurore Fass**, and Zakir Durumeric. Cloud Watching: Understanding Attacks Against Cloud-Hosted Services. In *ACM Internet Measurement Conference (IMC)*, 2023.
Acceptance rate: 25% (52 / 208 full research papers).

* Kimberly Ruth, **Aurore Fass**, Jonathan J. Azose, Mark Pearson, Emma Thomas, Caitlin Sadowski, and Zakir Durumeric. A World Wide View of Browsing the World Wide Web. In *ACM Internet Measurement Conference (IMC)*, 2022.
Acceptance rate: 26% (56 / 212 full research papers).

* **Aurore Fass**, Dolière Francis Somé, Michael Backes, and Ben Stock. DoubleX: Statically Detecting Vulnerable Data Flows in Browser Extensions at Scale. In *ACM CCS*, 2021. Code repository: `https://github.com/Aurore54F/DoubleX`.
Acceptance rate: 23% (131 / 564 full research papers, May cycle).

Marvin Moog, Markus Demmel, Michael Backes, and **Aurore Fass**. Statically Detecting JavaScript Obfuscation and Minification Techniques in the Wild. In *IEEE/IFIP Dependable Systems and Networks (DSN)*, 2021. Code repository: `https://github.com/MarM15/js-transformations`.
Acceptance rate: 16% (48 / 295 full research papers).

* **Aurore Fass**, Michael Backes, and Ben Stock. HideNoSeek: Camouflaging Malicious JavaScript in Benign ASTs. In *ACM CCS*, 2019. Code repository: `https://github.com/Aurore54F/HideNoSeek`.
Acceptance rate: 14% (32 / 225 full research papers, February cycle).

Aurore Fass, Michael Backes, and Ben Stock. JStap: A Static Pre-Filter for Malicious JavaScript Detection. In *ACSAC*, 2019. Code repository: `https://github.com/Aurore54F/JStap`.
Acceptance rate: 23% (60 / 266 full research papers).

Aurore Fass, Robert P. Krawczyk, Michael Backes, and Ben Stock. JaSt: Fully Syntactic Detection of Malicious (Obfuscated) JavaScript. In *DIMVA*, 2018. Code repository: `https://github.com/Aurore54F/JaSt`.
Acceptance rate: 32% (18 / 56 full research papers).

The publications are listed in reverse-chronological order. I marked the five most important ones with an *.

Community Services

Organizing Role USENIX Security Vice Chair 2027, USENIX Security **Artifact Evaluation Committee Co-Chair** 2026 & 2025, **Associate Editor** of the ACM Transactions on Security and Privacy (TOPS) 2024 & 2025, ACM CCS **Workshop General Co-Chair** 2024, MADWeb (workshop co-located with NDSS) 2024 & 2023 **PC Co-Chair** and MADWeb 2025– **Steering Committee**

PC Member ACM CCS 2026–2021, ACSAC 2026–2023, USENIX Security 2025 & 2024, IEEE EuroS&P 2024 & 2023, IEEE S&P 2023, TheWebConf 2023 & 2022, ARES 2023 & 2022, SecWeb 2024–2021

Doctoral Committee
Gilles Mertens (Ph.D., Computer Science, Université Grenoble Alpes, March 2026)
Jean Luc Intumwayase (Ph.D., Computer Science, Université de Lille, December 2024)
Romain Fouquet (Ph.D., Computer Science, Université de Lille, May 2023)

Project Proposal Reviewed projects for several European funding organizations (2023)

Artifact Committee USENIX Security 2021, ACSAC 2018

External Reviewer IEEE S&P 2024, TWEB 2024, ESORICS 2023, ICCCN 2023, NDSS 2022–2020, USENIX Security 2022–2020, IEEE EuroS&P 2019, ACSAC 2019 & 2018, ACM CCS 2018

Misc IMC Travel Grants 2023, CISPA Faculty Hiring Committee 2021

Teaching

WS 2026 **Machine Learning for Security: Detecting Malicious Code** (EUR DS4H, Université Côte d'Azur, FR)
- 24 hours, MSc students

WS 2026 **Applied Machine Learning for Security: Malware Detection** (Polytech Nice, Université Côte d'Azur, FR)
- 12 hours lecture + 16 hours exercices, MSc students (M2)

SS 2025 **The Web Security Seminar** (Saarland University, DE)

WS 2024–2025 **Guest Lectures** on Web Security & Security of Browser Extensions:
- **University of Bologna**, IT
- **University of Modena and Reggio Emilia**, IT

WS 2024–2025 **The Web Security Seminar** (Saarland University, DE)

SS 2024 **The Web Security Seminar** (Saarland University, DE)

WS 2023–2024 **The Web Security Seminar** (Saarland University, DE)
- Malicious JavaScript Analysis
- Beyond Malicious Extensions: How can Extensions put User Security & Privacy at Risk?
- User Browsing Behavior vs. Top Lists

WS 2020–2021 **Lecturer at TELECOM Nancy** (Université de Lorraine, FR)
- Browser Extensions: Architecture and Security Consideration (lectures and practicals for MSc students)

WS 2019–2020 **Seminar: Joint Advances in Web Security** (Saarland University, DE)
- Browser Extensions: Security and Vulnerabilities
- Overview of Malicious JavaScript Detection Techniques and Attacks

WS 2018–2019 **Seminar: Joint Advances in Web Security** (Saarland University, DE)
- Overview of Malicious JavaScript Detection Techniques
- Cryptojacking: Definition, Detection, and Dimensions

Advising and Mentoring

I left CISPA (Germany) at the end of November 2025 and joined Inria in December 2025. My PhD students and research assistant are still physically located at CISPA and Saarland University:

Ph.D. Students

Apr 2024– **Valentino Dalla Valle** – *Browser Extension Security* → *TWEB 2026 + papers under submission*, Saarland University & CISPA

Dec 2023– **Dominic Troppmann** – *Type Checks* → *ASE 2024 + paper under submission*, co-supervised with Cristian-Alexandru Staicu, Saarland University & CISPA

Research Assistant

Dec 2024– **Laith Alhelwane** (MSc student) – *Browser Extension Security*, Saarland University

Alumni

2025 **Abdullah Hassan Chaudhry** (MSc student) – *Browser Extension Security* → *MADWeb 2026*, Saarland University

Huda Dawoud (she was a PhD student in my group at CISPA until I left in November 2025, now she is working with Sven Bugiel) – *Browser Extension Analysis*, CISPA

2023–2024 **Ben Rosenzweig** (BSc thesis) – *Machine Learning-Based Approach for Detecting Malicious Browser Extensions* → *TWEB 2026*, Saarland University

2022–2023 **Sheryl Hsu** (BSc student → Research Engineer OpenAI) – *Browser Extension Security* → *AsiaCCS 2024*, Stanford University

Manda Tran (MSc student → Ph.D. student UCLA) – *Browser Extension Security* → *AsiaCCS 2024*, Stanford University

Liz Izhikevich (Ph.D. student of Zakir Durumeric → Assistant Professor UCLA) – *Internet Scanning* → *IMC 2023*, Stanford University

2021–2023 **Shubham Agarwal** (Ph.D. student of Ben Stock → PostDoc MPI-SP) – *Browser Extension Security* → *CCS 2024*, Saarland University & CISPA

Kimberly Ruth (Ph.D. student of Zakir Durumeric) – *Web Browsing Behavior* → *IMC 2022 + CHI 2025*, Stanford University

2022 **Mark Tran** (BSc student) – *Browser Extension Fingerprinting*, Stanford University

Vrushank Gunjur (BSc student) – *Over-Privileged Extensions*, Stanford University

Nahum Maru (BSc student) – *Browser Extension Crawler*, Stanford University

Fengchen (Maggie) Gong (MSc student → Ph.D. student Princeton) – *Fingerprinting*, Stanford University

2021 **Liana Patel** (Ph.D. student of Zakir Durumeric) – *Crawler*, Stanford University

Luca Pistor & **Nathan Bhak** (BSc students) – *Exam Software Security*, Stanford University

Paul Szymanski (BSc thesis) – *A Study of State-of-the-Art Call Graph Creation Approaches for JavaScript*, with Cristian-Alexandru Staicu, Saarland University & CISPA

2020 **Anne Christin Deutschen** & **Luc Seyler** (BSc students) – *Browser Extension Vulnerability*, with Dolière Francis Somé, Saarland University & CISPA

2019–2020 **Marvin Moog** & **Markus Demmel** (BSc students) – *Analysis of JavaScript Obfuscation Techniques* → *DSN 2021*, Saarland University & CISPA

2019 **Maximilian Zöllner** & **Niklas Kempf** (BSc students) – *Intelligent Fuzzing System for JavaScript*, Saarland University & CISPA

2018 **Nils Glörfeld** (BSc student) – *Malicious JavaScript Deobfuscation*, Saarland University & CISPA

Dennis Salzmann (BSc student) – *Malicious JavaScript Detection*, Saarland University & CISPA

Invited Talks

On the Security and Privacy Risks of Browser Extensions

Sep 2026 Google / Chrome Security Presentation. Remote.

Jul 2026 Dagstuhl Seminar 26291. Wadern, Germany.

Jun 2026 Seminar@SystemX. Remote.

Jun 2026 Plugin Seminar at the Inria Centre at Université Grenoble Alpes. Montbonnot-Saint-Martin, France (recording to come...).

Jun 2026 Keynote at APVP 2026. Trégastel, France.

May 2026 Keynote at RESSI 2026. Clervaux, Luxembourg.

Mar 2026 PrivaSki Seminar. Correncon en Vercors, France.

Jan 2026 Keynote at the GDR RSD/ASF Winter School on Distributed Systems&Networks. Pleynet, France.

Browser Extension (In)Security

Jan 2025 Seminar at Inria Sophia Antipolis. Sophia Antipolis, France.

Dec 2024 Seminar at Inria Lille. Lille, France.

Jun 2024 GDR Information Security. Rennes, France.

Dos and Don'ts of Reviewing

Nov 2024 Keynote at the Winter School, WinterHack 2024. Bochum, Germany.

DoubleX: Statically Detecting Vulnerable Data Flows in Browser Extensions

Nov 2023 Workshop at Inria. Paris, France.

Jul 2022 Berkeley Security Seminar. Berkeley, CA, U.S.

May 2022 RuhrSec. Bochum, Germany (extended version).

Apr 2022 Stanford Computer Forum – Security Workshop. Stanford, CA, U.S.

Nov 2021 Stanford Security Lunch. Stanford, CA, U.S.

Studying JavaScript Security Through Static Analysis

Apr 2024 PEPR Cyber – Project DefMal Webinar (France). Remote (extended version).

Mar 2022 Palo Alto Networks (CA, U.S.). Remote (extended version).

Jun 2021 Webinar at Inria Lille (France). Remote.

Statically Analyzing Malicious JavaScript in the Wild

Mar 2021 Webinar at LORIA (France). Remote.

Dec 2020 BINSEC Webinar at CEA (France). Remote.

HideNoSeek: Camouflaging Malicious JavaScript in Benign ASTs

May 2020 RuhrSec (Germany). Remote (extended version).

Mar 2019 Grande Region Security and Reliability Day (GRSRD). Nancy, France.

Feb 2019 MADWeb. San Diego, CA, U.S.

JaSt: Fully Syntactic Detection of Malicious (Obfuscated) JavaScript

Nov 2018 Blackhoodie. Berlin, Germany.
Jun 2018 Malware Meeting at LORIA. Nancy, France.
Mar 2018 Grande Region Security and Reliability Day (GRSRD). Saarbrücken, Germany.

Publicly Available Software

All the software I developed is publicly available on my GitHub account:

static-pdg-js	Static analysis of JavaScript code (AST, control & data flows, pointer analysis)
DoubleX	Static browser extension analyzer: detection of suspicious external data flows
HideNoSeek	Static analyzer to detect syntactic clones in JavaScript inputs
JStap	Static and modular malicious JavaScript detector
JaSt	Static malicious JavaScript detector
reimpl-cujo	Reimplementation of Cujo, static malicious JavaScript detector
reimpl-zozzle	Reimplementation of Zozzle, static malicious JavaScript detector

Additional Skills – Languages

French	Mother tongue	
English	Trilingual proficiency	*TOEIC score: 910 (2014); lived in the U.S. 2021–2023*
German	Trilingual proficiency	*C1 Certificate (2016); lived in Germany 2017–2021 & 2023–2025*